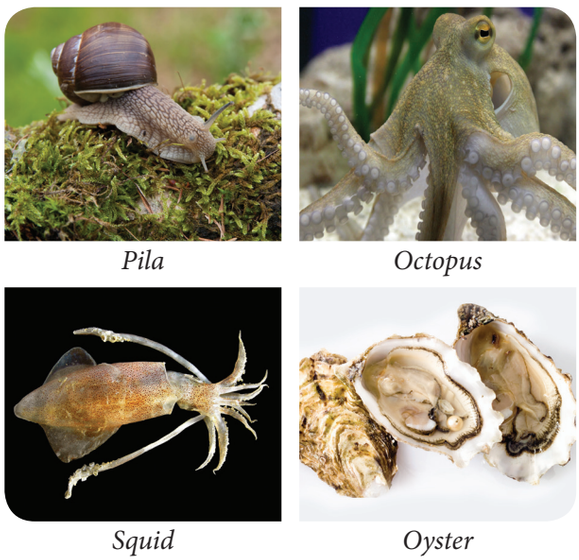
respiratory pigment. They are dioecious and oviparous. Development is indirect with a **veliger larva** (a modified trochophore larva).

Examples: *Pila* (Apple snail), *Lamellidens* (Mussel), *Pinctada* (Pearl oyster), *Sepia* (Cuttle fish), *Loligo* (Squid), *Octopus* (Devil fish) (Figure 2.16).



*Pila* *Octopus*

*Squid* *Oyster*

**Figure 2.16** Examples of Mollusca




**Marbled Cone Snail**
(*Conus marmoreus*)

This cone-shaped snail can deliver dangerous venom which may result in vision loss, respiratory failure, muscle paralysis and eventually death. There is no anti-venom available.

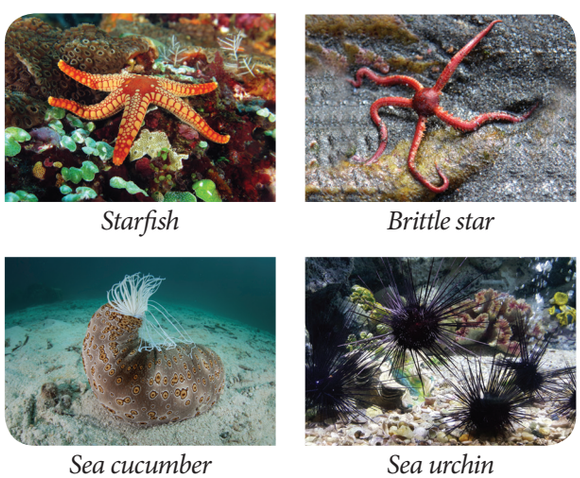


## 2.3.9 Phylum Echinodermata

(G. *Echinos* – spiny; *dermos* –skin)

All Echinoderms are marine animals. The adults are **radially symmetrical** but the larvae are **bilaterally symmetrical**. These animals have a mesodermal endoskeleton of calcareous ossicles and hence the name Echinodermata (spiny skin). They are exclusively marine with organ system level of organisation. The most distinctive feature of echinoderms is the presence of the **water vascular system** or a**mbulacral system** with **tube feet** or **podia**, which helps in locomotion, capture and transport of food and respiration. The digestive system is complete with mouth on ventral side and anus on the dorsal side. Excretory organs are absent. The nervous system and sensory organs are poorly developed. The circulatory system is open type without heart and blood vessels. Sexes are separate. Reproduction is sexual and fertilization is external. Development is indirect with free swimming bilaterally symmetrical larval forms. Some echinoderms exhibit autotomy with remarkable powers of **regeneration.** e.g. Star fish. (Figure 2.17)

Examples: *Asterias* (Starfish or Sea star), *Echinus* (Sea-urchin), *Antedon* (Sea-lily), *Cucumaria* (Sea-cucumber), *Ophiura* (Brittle star)



*Starfish* *Brittle star*

*Sea cucumber* *Sea urchin*

**Figure 2.17** Examples of Echinodermata



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